



INDUSTRIAL ENGINEERING & AUTOMATION PRACTICE

Engineering Capability Brochure

EXECUTIVE PRACTICE MANDATES & ADVISORY FOCUS

"Precision mechanical design, CAD/CAE simulation, Special Purpose Machine (SPM) building, PLC industrial automation, and rapid prototyping for Indian manufacturing enterprises."

CORE PRACTICE AREAS INCLUDED IN THIS BROCHURE

- **Special Purpose Machines (SPM)** Turnkey assembly, testing, inspection, and packaging machinery designed for target cycle times.
- **Industrial Automation & Robotics** PLC programming (Siemens, Allen-Bradley, Mitsubishi), SCADA, servo control, and vision inspection.
- **CAD, FEA & Reverse Engineering** SolidWorks/Inventor 3D modeling, FEA structural stress analysis, DFM reviews, and 3D laser scanning.

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01 - Turnkey Special Purpose Machines (SPM)

Custom automated machine design and manufacturing to replace manual bottlenecks and multiply throughput.

Custom Machine Building Categories

- Automated Component Assembly Stations with pneumatic, hydraulic, and servo integration
- End-of-Line (EOL) Testing Rigs: Leak testing, pressure endurance, torque verification
- Multi-Station Rotary Indexing Table Machines for high-speed repetitive manufacturing
- Automated Part Feeding & Sorting Systems (bowl feeders, linear tracks, pick-and-place units)

Manufacturing Standards & Components

- Pneumatics & Actuators: Festo, SMC, and Janatics precision valves and cylinders
- Linear Motion: THK and Hiwin guideways, ground ballscrews, and heavy-duty slide units
- Sensors & Safety: Keyence, Sick, and Omron optical sensors, light curtains, and interlocks
- Structural Frames: Precision laser-cut mild steel, stress-relieved weldments, and anodized aluminum extrusions

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02 - Industrial Automation, PLC & SCADA

Complete electrical architecture, control panel fabrication, and intelligent automation.

Control System Capabilities

- PLC Programming: Siemens S7-1200/1500, Allen-Bradley Micro800/CompactLogix, Mitsubishi, Delta
- HMI Interface Design: User-friendly touchscreen interfaces with recipe management and diagnostics
- Servo & Motion Synchronization: Multi-axis servo drives for precise positioning and tension control
- SCADA & Factory Data Acquisition: Real-time OEE tracking, downtime analytics, and MES integration

Quality & Machine Vision Integration

- Industrial Machine Vision Systems (Cognex, Keyence) for sub-millimeter defect detection
- Automated 100% Dimensional Inspection Stations eliminating human inspection errors
- Barcoding, 2D DataMatrix marking, and traceability integration directly into production databases

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03 - Precision Mechanical CAD & Simulation

Robust digital engineering eliminating design flaws before metal is cut.

Design & Modeling Practice

- Parametric 3D CAD Modeling in SolidWorks, Autodesk Inventor, and Siemens NX
- Detailed Manufacturing Drawings: ASME Y14.5 compliant Geometric Dimensioning & Tolerancing (GD&T)
- Design for Manufacturing & Assembly (DFM/DFA) reviews to minimize CNC machining costs
- BOM Generation: Complete indented bills of materials with material and standard hardware specs

Finite Element Analysis (FEA)

- Static Structural Stress Analysis: Verifying safety factors under maximum operational loads
- Dynamic & Vibration Analysis: Eliminating harmonic resonance and premature fatigue failures
- Thermal Simulation: Evaluating heat dissipation in industrial enclosures and drive mechanisms
- Topology Optimization: Strategic lightweighting of structural castings and machined brackets

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04 - Reverse Engineering & Rapid Prototyping

Digitizing obsolete imported components and producing functional prototypes rapidly.

Reverse Engineering Workflow

- High-Precision 3D Optical & Laser Scanning of legacy tooling and worn machine parts
- CAD Reconstruction: Converting point clouds and mesh data into parametric STEP/IGES models
- Material Composition & Hardness Testing: Identifying exact alloy metallurgy and heat treatment
- Domestic Import Substitution: Engineering domestic alternatives to expensive OEM spare parts

Rapid Prototyping & Verification

- Industrial 3D Printing (FDM, SLA, SLS) in nylon, ABS, and polycarbonate for form/fit checks
- Rapid CNC Machining: 3-axis and 5-axis prototype fabrication in steel, aluminum, and brass
- Functional Sheet Metal Prototypes: Laser cutting, CNC bending, and rapid TIG/MIG welding

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05 - Fabrication, In-House Assembly & Trials

High-precision manufacturing infrastructure ensuring built-to-print perfection.

Manufacturing Ecosystem

- CNC Milling & Turning: High-precision machining centers delivering +/- 10 micron tolerances
- Precision Grinding: Surface grinding and cylindrical grinding for hardened spindle and slide surfaces
- Stress Relieving & Heat Treatment: Post-weld normalization, induction hardening, and nitriding
- Dedicated Clean Assembly Bay: Controlled mechanical and pneumatic assembly with calibrated torque tools

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06 - Testing, FAT, Site Commissioning & AMC

Rigorous dry and wet trials before machine sign-off and factory shipment.

Validation & Commissioning Protocol

- Factory Acceptance Test (FAT): Continuous 8-hour uninterrupted dry and production component trials
- Cycle Time & CPK Verification: Establishing machine capability index (CPK > 1.67) on critical dimensions
- Site Installation & Commissioning: Doorstep machine levelling, utility hookup, and trial production
- Operator & Maintenance Training: Complete technical documentation, electrical schematics, and hands-on training

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07 - Warranty, Spare Parts & Maintenance Contracts

Standing technical support to guarantee zero unscheduled factory downtime.

Standing Engineering Support

- Comprehensive 12-Month Mechanical & Electrical Machine Warranty
- Guaranteed Spare Parts Availability: Recommended 2-year consumable spares pack supplied with delivery
- Annual Maintenance Contracts (AMC): Scheduled quarterly preventative audits and recalibration
- On-Site Breakdown Support: Rapid-response engineering technicians dispatched within 24 hours

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08 - Commissioning Your Engineering Project

How to engage Jyot Engineering for machine building and design mandates.

Engagement Process

- Step 1: Share component drawings (2D/3D), cycle time targets, and batch specifications
- Step 2: Joint concept brainstorm & DFM review with our Senior Machine Design Team
- Step 3: Receive 3D concept layout, detailed technical proposal, and commercial timeline
- Direct Engineering Desk: +91 95374 30101 * Email: jyotenterpriseofficial@gmail.com